

multiple applications in a single OS instance. They are lightweight, portable and offer a more efficient way to deploy and manage applications than traditional virtual machines.

Here are some reasons why Linux containers are gaining popularity.

**Resource efficiency:** Containers share the operating system kernel, which means they use fewer system resources than virtual machines. This makes them ideal for running multiple applications on a single server or instance.

**Portability:** They are portable and can be easily moved from one system to another. This makes it easy to deploy applications across different environments, such as development, testing, and production.

**Consistency:** Containers ensure that applications run consistently across different systems. This is especially important in complex multi-cloud environments where applications need to run on different platforms and operating systems.

**Security:** They provide robust isolation between applications, which reduces the risk of security threats. With the right security measures in place, containers can be more secure than traditional virtual machines.

**Speed:** Containers provide fast application deployment and are quick to start up or shut down. This means that developers can test and deploy new features faster, improving their agility and time to market.

## Understanding containers

Containerisation technology has gained a lot of popularity in recent years, with Docker being at the forefront of the movement. It allows developers to create, run, and manage applications in containers that are independent of the underlying infrastructure. This makes the process of deploying applications across multiple environments, such as development, testing, and production, a lot easier.

Containers are isolated from each other, ensuring that applications running inside them do not interfere with one another. Furthermore, they provide an additional layer of security that can prevent the spread of vulnerabilities and attacks across applications.

Each container runs in its own name space, meaning that even if one container is compromised, the threat is limited to only that container and cannot affect other applications on the same machine.

Containers are also highly scalable, as they can be easily replicated and deployed across multiple machines. This makes them ideal for microservices architecture, where applications are broken down into smaller, more manageable components that can be scaled up or down as needed.

### Types of containers

There are several types of Linux containers, including:

- Docker
- LXC (Linux Containers)
- Kubernetes
- OpenVZ
- systemd-nspawn
- Rkt (Rocket)
- Snappy

## Popular containerisation platforms

Containerisation platforms provide users with an efficient, scalable, secure, and cost-effective solution for their application deployment needs. There are several popular containerisation platforms in the market that offer a range of features and functionalities. Here, we discuss three of the most widely used ones.

**Docker:** Docker is the most widely used containerisation platform that provides a complete solution for container creation, deployment, and management. It is an open source platform that can run on any operating system, including Windows, Linux, and Mac. Docker allows developers to create containerised applications and deploy them on any infrastructure, including cloud, on-premise, and hybrid environments. The platform also offers a wide range of tools and features, such as Docker Swarm for orchestration, Docker Compose for multi-container applications, and Docker Hub for the distribution of containerised applications.

**Kubernetes:** Kubernetes is an open source containerisation platform for automating the deployment, scaling, and management of containerised applications. It is designed to work with multiple containerisation platforms, such as Docker, and supports a range of containerisation tools and platforms. Kubernetes provides container orchestration, automatic scaling, and self-healing mechanisms for containerised applications, making it a popular choice for enterprises. The platform also offers a range of features, such as load balancing, fault tolerance, and automatic deployment rollbacks.

**OpenShift:** OpenShift is a container application platform built on top of Kubernetes and Docker. It is a container management platform that automates the deployment, scaling, and management of containerised applications. OpenShift offers developers a complete solution for building, testing, and deploying containerised applications on any infrastructure. The platform provides support for multiple languages, frameworks, and databases, and it also offers scalability and high availability for enterprise-class applications. OpenShift provides an integrated environment for developers to manage the whole application life cycle, from code to deployment.